ENBC321/BIOE658I: Homework #7

Due: Monday 04/13/26 by 5:00 PM, beginning of the class

Submit your homework to ELMS, including Python code and the dataset

Overall points: 100

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "`problem1_fullname.ipynb`". Start with the problem number and followed by your full name, e.g., "**problem1_jimmyazarnoosh**".

Note: Run all cells and ensure there are no syntax errors. All results must be displayed in the notebook. (Syntax errors = zero points).

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of ZERO. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- Python code files in ***.ipynb**
- Your answers to the questions in PDF format

Problem 1. K-Means Clustering (50 points)

A hospital is analyzing the health profiles of patients to identify subgroups with similar clinical features, which may correspond to different disease subtypes or treatment needs. As a data scientist working with the medical research team, your task is to use K-means clustering to analyze the patient data. The dataset contains anonymized data of 120 patients, each with the following features:

- **Age:** Patient's age (in years)
- **BMI:** Body Mass Index
- **Blood Pressure:** Systolic blood pressure (mmHg)
- **Glucose Level:** Fasting blood glucose (mg/dL)
- **Insulin Level:** Fasting insulin level (μU/mL)

Tasks:

- Load the dataset from the provided CSV file: **patients_data.csv**.
- Perform **two separate** clustering analyses using different feature combinations:
  1) Use **Age** and **Blood Pressure**.
  2) Use **Age** and **Glucose Level**.
- Visualize the raw data using a 2D scatter plot.
- Normalize the features using standard scaling (mean=0, std=1).
- Apply K-Means clustering using K-Means++ initialization on each selected feature pair. (10 points)
- For each analysis: (15 points)
  - Use the **Elbow Method** to determine the appropriate number of clusters (k = 2 to 10).
  - Use **Silhouette Analysis** to validate the number of clusters and interpret clustering quality.
- For each feature pair, plot the clustered data on a 2D scatter plot with different colors for each cluster and clearly marked cluster centers. (10 points)

Conceptual questions:

1. Why is feature scaling important before applying K-means clustering? Hint: Think about how K-means uses distance in feature space. (5 points)
2. If you run K-means multiple times with the same data and different initial seeds, can you get different results? Why? How does K-means++ address this? (5 points)
3. What limitations does K-means clustering have, and in what situations might you prefer another clustering method like DBSCAN or hierarchical clustering? (5 points)

Problem 2. Hierarchical Clustering (50 points)

In this problem, you will use **Agglomerative Hierarchical Clustering** to analyze the same dataset in problem 1, a dataset containing 120 anonymized patient records.

Tasks:

- Load the dataset from the provided CSV file: **patients_data.csv**.
- Perform **two separate** clustering analyses using different feature combinations:
  1) Use **Age** and **Insulin Level**.
  2) Use **Age** and **BMI**.
- Visualize the raw data using a 2D scatter plot.
- Normalize the features using standard scaling (mean=0, std=1).
- Use Agglomerative Clustering with the following settings:
  - Linkage methods to use: Ward
  - Distance metric: Euclidean
- Generate and plot the dendrogram using `scipy.cluster.hierarchy.dendrogram`. (10 points)
- Based on the dendrogram, identify a reasonable number of clusters (use visual inspection and your judgment). (10 points)
- Create a 2D scatter plot of the clustered data with different colors representing different clusters. (5 points)
- For each feature pair, plot the dendrogram and clustered data on a 2D scatter plot with different colors for each cluster and clearly marked cluster centers. (10 points)

Conceptual questions:

1. How does agglomerative hierarchical clustering differ from K-means clustering in terms of approach and assumptions? (5 points)
2. Why might hierarchical clustering be preferable to K-means in certain applications, particularly with patient data? (5 points)
3. What challenges or limitations might arise when using hierarchical clustering on higher-dimensional data (e.g., using all 5 features together)? (5 points)